PRATHAM DESAI

B.TECH CSE GRADUATE

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13, Umiyanagar Society, Near Narmada Canal, Panchvati, Gorwa, Vadodara - 390016

EDUCATION

BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE ENGINEERING : 6.92 CGPA | 69.2%

GSFC University

2022-2026

CLASS 12th : 63% (CBSE)

Kendriya Vidyalaya No.2
E.M.E Baroda

2022

CLASS 10th : 72% (CBSE)

Kendriya Vidyalaya No.2
E.M.E Baroda

2020

SKILLS

- Strong Communication Skills
- Proficient in Microsoft Office Suite
- Solid Technical Knowledge
- Foundational Programming Skills
- Video Editing Skills
- Prompt Engineering Proficiency
- Basic Photography/Videography Skills
- Effective Teaching and Leadership Abilities

PROFILE

Date of Birth: 16th September 2004

Gender: Male

I am a passionate tech enthusiast with a deep interest in computer science and technology. Along with my technical knowledge, I possess strong communication and leadership skills. I enjoy helping others learn and grow. I'm always eager to learn new things and take on challenges in the tech world.

LANGUAGES

- Gujarati (Native): proficiency in reading, writing, speaking and comprehension.
- Hindi: proficiency in reading, writing, speaking, and comprehension
- English: proficiency in reading, writing, speaking, and comprehension

ACADEMIC PROFILE

	SGPA	CGPA
1st Semester:	6.00	6.00
2nd Semester:	6.88	6.43
3rd Semester:	5.96	6.27
4th Semester:	6.08	6.23
5th Semester:	7.14	6.39
6th Semester:	7.37	6.58
7th Semester:	7.15	6.65
8th Semester:	10.00	6.92

EXPERIENCE

1. **ACADEMIC ASSOCIATE: PYTHON PROGRAMMING**

GSFC University, Vadodara, Gujarat
July 2025 - October 2025

- Conducted lab lectures for BTech CSE 3rd semester students, teaching Python programming fundamentals
- Provided supplementary learning materials to enhance student comprehension
- Maintained attendance records and academic progress reports for approximately 28 students
- Prepared students for end-semester practical examinations, focusing on programming concepts and problem-solving techniques
- Provided individualized support to students struggling with programming concepts & Served as an accessible mentor, bridging the gap between faculty and students

2. **ACADEMIC ASSOCIATE: PYTHON PROGRAMMING**

GSFC University, Vadodara, Gujarat
January 2025 - April 2025

- Conducted lab lectures for BTech FEHS 2nd semester students, teaching Python programming fundamentals
- Provided supplementary learning materials to enhance student comprehension
- Maintained attendance records and academic progress reports for approximately 35 students
- Prepared students for end-semester practical examinations, focusing on programming concepts and problem-solving techniques
- Provided individualized support to students struggling with programming concepts & Served as an accessible mentor, bridging the gap between faculty and students

3. **ACADEMIC ASSOCIATE: C PROGRAMMING**

GSFC University, Vadodara, Gujarat
August 2024 - November 2024

- Conducted lab lectures for BSc Data Science 1st semester students, teaching C programming language fundamentals
- Provided supplementary learning materials to enhance student comprehension
- Maintained attendance records and academic progress reports for approximately 45 students
- Prepared students for end-semester practical examinations, focusing on programming concepts and problem-solving techniques
- Provided individualized support to students struggling with programming concepts & Served as an accessible mentor, bridging the gap between faculty and students

INTERNSHIP

1.

CYBERSECURITY INTERN (8^{TH} SEM)

Tata Consultancy Services (TCS) Garima Park, Gandhinagar, Gujarat
5^{th} February 2026 - 5^{th} June 2026

- Designed and developed **CipherLens**, a comprehensive web-based platform for generating and analyzing Cryptographic Bills of Materials (CBOM) across source code, Git repositories, Docker images, and Kubernetes clusters.
- Architected a highly concurrent backend using Python and FastAPI, implementing a custom ProcessPoolExecutor-based scheduling engine with IPC queues and WebSockets for real-time log streaming of isolated background scans.
- Built a robust security validation engine to evaluate discovered cryptographic components against NIST SP 800-131A, CERT-In, and ETSI standards, calculating algorithmic risk levels and PQC (Post-Quantum Cryptography) readiness scores.
- Engineered a scalable data layer utilizing PostgreSQL for persistent scan history and integrated OpenLDAP with JWT authentication to enforce strict Role-Based Access Control (RBAC) in enterprise, multi-user deployments.
- Implemented a high-performance dual-scanner pipeline integrating CycloneDX `cdxgen` with a custom parallel regex fallback scanner, successfully optimizing scan duration and minimizing resource contention.
- Developed an interactive frontend dashboard using React, Material UI (MUI), and Recharts to visualize security metrics, facilitate side-by-side scan comparisons, and generate dynamic reports in PDF, HTML, CSV, and JSON formats.
- Worked with enterprise development workflows involving access provisioning, RFC and ticket management processes, software installation approvals, environment configuration, and corporate governance practices within large-scale organizations.

Reference:

1. Mr. Ryan Mathew - Project Lead at PayGURU - akshay.bhatt@tcs.com
2. Mr. Bhaumik Machhi - Faculty Mentor at GSFC University - bhaumik.machhi@gsfcuniversity.ac.in

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PYTHON DEVELOPER INTERN (7TH SEM)

PayGURU Technosoft Pvt Ltd, Vadodara, Gujarat
1st December 2025 - 10th January 2026

- Designed and developed multiple production-grade software systems including a Visitor Management System, WhatsApp Business CRM, CCTV-based Face Recognition System, and security automation utilities.
- Architected scalable backend services using Python, Node.js, Flask, REST APIs, and multi-threading, focusing on performance, reliability, and real-world deployment constraints.
- Built SaaS-ready multi-tenant architectures with strict data isolation, dynamic white-labeling, and centralized configuration to support multiple enterprise clients from a single instance.
- Engineered real-time systems integrating live CCTV feeds, RTSP/ISAPI streams, WhatsApp webhooks, and email notifications to automate attendance, visitor approvals, and marketing workflows.
- Implemented high-performance data pipelines including GPU-accelerated face recognition, FAISS-based vector search, bulk contact ingestion, and optimized database persistence.
- Developed secure automation tools for IoT and enterprise environments, featuring session-based access control, encrypted credential management, automated password rotation, and fallback browser automation.
- Collaborated closely with product and business teams to convert operational requirements into deployable software solutions, reducing manual effort and improving system reliability.

Key Projects:

- Visitor Management System (PookiePass)
- Enterprise WhatsApp Marketing Bot
- CCTV Face Recognition & Attendance System
- Session based Access Control python utility

Reference:

- Mr. Ryan Mathew - Project Lead at PayGURU - +91 98795 39606
- Ms, Divya Kushwah - Faculty Mentor at GSFC University - +91 99982 72375

3. PYTHON DEVELOPER INTERN (6TH SEM)

PayGURU Technosoft Pvt Ltd, Vadodara, Gujarat

9th June 2025 - 4th July 2025

- Collaborated in a two-member team to design, develop, and deploy "PookieTrak," a cross-platform desktop application for real-time monitoring of software services and network-connected devices.
- Developed modern, user-friendly GUIs using PyQt6 and implemented robust backend logic for process and device tracking.
- Integrated multi-threading, automated reporting, and email notification systems to enhance operational efficiency and reliability.
- Conducted systematic testing, debugging, and performance optimization to ensure seamless functionality across Windows and Linux environments.
- Managed data storage and reporting using JSON and CSV formats, ensuring data integrity and security.
- Documented code, prepared user guides, and participated in regular code reviews and team meetings.
- Gained hands-on experience in the full software development lifecycle, from requirements gathering to deployment and maintenance.

Reference:

1. Mr. Ryan Mathew - Project Lead at PayGURU - +91 98795 39606
2. Ms, Divya Kushwah - Faculty Mentor at GSFC University - +91 99982 72375

4. MERN STACK DEVELOPER INTERN (5TH SEM)

Sannibh Technologies, Vadodara, Gujarat

2nd December 2024 - 3rd January 2025

- Collaborated in a two-person team to design and develop a full-stack ERP software ("Brass ERP") for brass part manufacturing using the MERN stack (MySQL, Express.js, React, Node.js).
- Built secure authentication, client management, and dynamic dashboards, integrating RESTful APIs and state management (Zustand) for robust performance.
- Implemented responsive UI with TailwindCSS and Vite, ensuring a seamless user experience across modules like quotations, production, and sales.
- Worked in a hybrid environment, independently managing codebase and delivering features while reporting progress in-office.
- Gained hands-on experience in modern web development, database integration, and scalable application architecture.
- Enhanced problem-solving and communication skills by closely coordinating with a teammate and stakeholders to meet project requirements.

PROJECTS

1. **CIPHERLENS: CBOM ANALYZER TOOL (TCS INTERNSHIP)**

CipherLens (PODMAN VERSION)
2nd June 2026

Technologies: Python, FastAPI, React.js, Podman, Kubernetes (kubectl), SQLite, Server-Sent Events (SSE).

Description: Developed "CipherLens (Podman Version)," an automated cryptographic auditing platform focused on containerized workloads, capable of extracting and analyzing OCI container layers directly via Podman and querying active Kubernetes clusters.

Key Technical Highlights:

- Rootless Container Scanning: Integrated the Podman daemon via socket API to securely pull and unpack container image layers into tar archives for deep cryptographic analysis without requiring root administrator privileges.
- Real-Time Scan Observability: Built a highly responsive logging pipeline utilizing Server-Sent Events (SSE) to broadcast live stdout/stderr traces from parallel background workers (Podman, cdxgen) directly to the React frontend console.
- Kubernetes Cluster Discovery: Engineered a cluster discovery module using kubectl to dynamically map active namespaces and pods, automating the extraction and scanning of running container images for cluster-wide compliance tracking.
- Dual-Pipeline Aggregation: Engineered a robust backend merging system that correlates dependency analysis from CycloneDX with deep source code analysis from a custom regex scanner, dynamically deduplicating overlapping cryptographic signatures to produce an accurate CBOM.
- Comprehensive Regulatory Engine: Implemented an in-memory compliance compiler that evaluates discovered cryptographic assets against six international standards concurrently (including NIST, CERT-In, and CNSA 2.0), enabling instant risk categorization and mitigation recommendations.
- Transactional Data Persistence: Implemented a lightweight, highly reliable SQLite database layer utilizing ACID-compliant transactional execution blocks to guarantee data integrity and prevent record corruption during concurrent background scan updates.
- Cross-Environment Socket Mapping: Overcame critical runtime constraints by dynamically mapping the local rootless Podman socket to standard Docker daemon paths, ensuring seamless compatibility and execution for upstream CycloneDX container analysis tools.
- Multi-Format Reporting Pipeline: Engineered an integrated data export module that instantly compiles compliance findings into interactive HTML dashboards, vector-based PDF summaries, raw CSV spreadsheets, and strictly structured CycloneDX 1.6 JSON artifacts.

CipherLens (CLI VERSION)
13th May 2026

Technologies: Python, Click, SQLite, CycloneDX (cdxgen), Regex, ReportLab, Pydantic.
Description: Designed and developed "cipherCLI," a standalone, offline command-line utility for generating and validating CBOMs, integrating seamlessly into developer environments and CI/CD pipelines without requiring continuous cloud connectivity.
Key Technical Highlights:

- Dual-Mode Static Analysis: Implemented a highly optimized scanning pipeline running cdxgen subprocesses in parallel with a custom regex-based fallback scanner, efficiently capturing deeply nested and hardcoded cryptographic primitives.
- Localized Persistence & Reporting: Engineered a lightweight SQLite database layer with transactional integrity to record local scan history, while dynamically generating enriched JSON, CSV, HTML, and watermark-protected PDF reports.
- Seamless CI/CD Integration: Designed the utility with strict POSIX-compliant options, configurable timeout limits, and zero-configuration YAML rulesets, ensuring reliable compliance auditing within air-gapped corporate networks.

CipherLens (AVD VERSION)
1st May 2026

Technologies: Python, FastAPI, React.js, Material UI, PostgreSQL, OpenLDAP, CycloneDX, WebSockets.
Description: Designed and developed "CipherLens (AVD Version)," an enterprise-grade web platform for generating and analyzing Cryptographic Bills of Materials (CBOM) across source code, Git repositories, Docker images, and Kubernetes clusters.
Key Technical Highlights:

- Asynchronous Scanning Engine: Architected a robust ProcessPoolExecutor-based scheduling engine with IPC queues to handle resource-intensive asynchronous cdxgen scans without blocking the main FastAPI event loops.
- Multi-Standard Compliance Validation: Built a flexible YAML-based rules engine to validate discovered cryptographic components against NIST SP 800-131A, CERT-In, and ETSI standards, calculating an overall project risk score and Post-Quantum Cryptography (PQC) readiness.
- Scalable Enterprise Architecture: Engineered a secure data layer utilizing PostgreSQL for persistent scan history and integrated OpenLDAP with JWT authentication to enforce strict Role-Based Access Control (RBAC).

2. POOKIEPASS: VISITOR MANAGEMENT SYSTEM (7TH SEM)

10th January 2026

Technologies: React.js, Node.js, Express.js, MariaDB (Sequelize ORM), REST APIs, SMTP (Nodemailer), Multi-Tenancy Architecture.

Description: Designed and developed "PookiePass," a SaaS-ready Visitor Management System (VMS) for PayGURU Technosoft. The system replaces manual logbooks with a contactless digital check-in workflow, featuring multi-tenant data isolation to serve multiple client organizations simultaneously from a single instance.

Key Technical Highlights:

- SaaS Multi-Tenancy Architecture: Engineered a scalable single-instance, multi-tenant backend using custom middleware to resolve tenant slugs and enforce Row-Level Security, ensuring strict data isolation between different corporate clients.
- Real-Time Approval Workflow: Built a high-responsiveness notification system that instantly alerts hosts via HTML-rich emails and WhatsApp integration, featuring a polling-based "Waiting Room" UI that updates visitor status in real-time without page refreshes.
- Dynamic White-Labeling: Developed a flexible React frontend that dynamically adapts branding (logos, themes) based on the accessed company URL, providing a personalized and professional experience for each tenant organization.

3. WHATSAPP AUTOMATION BOT: FOR PAYGURU MARKETING (7TH SEM)

10th January 2026

Technologies: Python, Flask, Meta Graph API, SQLite, Multi-threading, PyQt Signals, JSON Configuration.

Description: Developed an intelligent WhatsApp Business CRM to automate marketing funnels for PayGuru Technosoft. The system replaces manual engagement with a 24/7 automated bot that manages product showcases and customer support workflows.

Key Technical Highlights:

- Scalable Data-Driven Engine: Refactored hardcoded messaging into a configuration-driven system, enabling non-developers to scale product portfolios without modifying the core codebase.
- High-Volume Data Processing: Built a bulk import feature on a custom SQLite layer to onboard thousands of client contacts efficiently for mass campaigns.
- Resilient Webhook Architecture: Engineered a multi-threaded Flask server with smart media caching, reducing Meta API latency by ~40% during high-traffic broadcasts.
- Real-Time Desktop Integration: Implemented an event-driven PyQt signal bridge to push new leads from cloud services to a local desktop GUI in real time.

4.

CCTV-BASED REAL-TIME FACE RECOGNITION & ATTENDANCE SYSTEM (7TH SEM)

20th December 2025

Technologies: Python, OpenCV, PyTorch (CUDA), InsightFace, DeepFace (ArcFace, MTCNN), FAISS, RTSP/ISAPI, NumPy, Pandas, Albumentations.

Description: Developed a production-ready face recognition and attendance system using live CCTV feeds, enabling real-time identity verification and automated employee logging. The system was designed for commercial deployment, directly integrating with Hikvision IP cameras and replacing manual or card-based attendance with a scalable biometric solution.

Key Technical Highlights:

- Built a GPU-accelerated real-time face recognition pipeline using InsightFace (ArcFace), achieving high accuracy under varying lighting, pose, and camera angles.
- Integrated FAISS vector search for fast and scalable face embedding matching across large identity databases with near-instant retrieval performance.
- Engineered a low-latency RTSP video processing pipeline with frame skipping, buffering, and multi-threading to ensure stable real-time performance.
- Implemented robust enrollment workflows with image quality validation and augmentation to improve recognition reliability in real-world environments.
- Designed identity stabilization, cooldown-based attendance logic, and automated CSV/database logging to prevent duplicate entries and reduce false positives during high traffic.

5.

SESSION BASED ACCESS CONTROL PYTHON UTILITY (7TH SEM)

7th December 2025

Technologies: Python, Multi-threading, REST APIs (ISAPI), MySQL, Selenium, AES Encryption.

Description: Designed and built a Python-based security automation tool to manage Hikvision IoT devices. The system eliminates manual administration by automating discovery, password rotation, and secure credential storage.

Key Technical Highlights:

- Intelligent Network Scanning: Engineered a high-concurrency scanner (50+ threads) to auto-discover devices and permanently track them using MAC/Serial fingerprinting, ensuring continuity even when IP addresses change.
- Secure Credential Management: Implemented an automated password rotation engine using custom ISAPI integration, ensuring all credentials are AES-512 encrypted and synchronized to a central MariaDB database.
- Hybrid Automation & Reliability: Developed a fault-tolerant architectures that seamlessly falls back to Selenium browser automation for legacy devices, ensuring 100% coverage across different firmware versions.

6. MAYABHEDI: THE ONE WHO SEES THROUGH DECEPTIONS (7TH SEM)

7th November 2025

Engineered a unified multi-domain cybersecurity application integrating AI-based surveillance detection, ML-driven phishing analysis, enterprise-grade cryptographic vaults, file steganography, network forensics, and system vulnerability assessment—all inside a modern PySide6 desktop interface.

Designed with 6 specialized security modules + 1 home dashboard, covering end-to-end cybersecurity domains:

- Password Guardian (Password Security & Breach Analysis):
  - Designed a military-grade password protection system using AES-256-GCM, Argon2id KDF, real-time password strength analysis (zxcvbn), and automatic breach monitoring for compromised credentials.
- Vault of Shadows (File Security & Steganography):
  - Built a secure file vault with authenticated encryption, integrated lossless compression, and developed a steganography engine that hides encrypted data inside images for covert communication.
- PhishEye (Web Security & Phishing Detection):
  - Developed an ML-powered phishing detection engine using scikit-learn, URL feature extraction, and probability-based scoring to classify malicious links with detailed threat explanations.
- Vulnerability Scanner (System & Network Hardening):
  - Implemented a scanning engine using Nmap, Shodan API, and system audit checks to detect misconfigurations, exposed ports, outdated services, and generated automated PDF security reports.
- Threat Monitor (Network Forensics & Threat Intelligence):
  - Created a real-time packet monitoring system using Scapy, with threat intelligence lookups, IP geolocation (Geopy, Folium), suspicious traffic alerts, and live cyber-threat mapping.
- Hidden Camera Detection (Computer Vision & Physical Security):
  - Trained and deployed a YOLOv8 model on a 5,500-image dataset to detect covert surveillance devices using CV-based lens detection, IR-glint scanning, and Wi-Fi-based IoT camera identification.
- PySide6 GUI (Modular, Multi-Threaded, Dark-Themed Interface):
- Designed a polished, responsive 7-page GUI (6 modules + home page) with multi-threaded task handling, modular architecture (UI/Utils separation), and a modern security-focused dark theme.

7. **SCANSAGE: ALL-IN-ONE DOCUMENT UTILITY FOR STUDENTS (7TH SEM)**

7th October 2025

ScanSage is an all-in-one student-focused Java Android utility app designed to make document management simple. It allows merging, splitting, converting, compressing, and scanning files easily in one place.

- Home Fragment (Dashboard & Navigation Hub)
  - Built a clean dashboard using Fragments, Navigation Components, and Material UI, offering quick access to all tools with smooth RecyclerView-driven navigation and async state handling.
- Merge PDF
  - Implemented efficient multi-PDF merging using iText7, drag-reorder support, and ExecutorService-based processing to avoid UI blockage during large merges.
- Split PDF
  - Developed flexible page extraction (specific pages, ranges, full split) using iText7 copyPagesTo(), ensuring accurate outputs and storage-safe file handling.
- Compress PDF
  - Added configurable compression tiers via WriterProperties, balancing file-size reduction with readability, with real-time compression ratio feedback.
- PDF Conversion (DOCX ↔ PDF ↔ HTML)
  - Enabled cross-format conversions using Apache POI, iText7, and HTML2PDF for text extraction, DOCX paragraph mapping, and CSS-aware HTML rendering.
- Scan to PDF (Camera + Vision Pipeline)
  - Engineered a full OpenCV pipeline: grayscale → Gaussian blur → Canny edges → contour detection → 4-corner approximation → perspective correction → enhancement. Integrated CameraX for high-resolution capture and built a reorderable page manager.
- PDF to Image
  - Implemented high-quality PDF rasterization via PdfRenderer, exporting to JPEG/PNG/WebP with 2× resolution scaling and optimized memory usage.
- Image Conversion
  - Created fast and reliable JPEG ↔ PNG ↔ WebP transformations using Bitmap APIs, with format validation and lifecycle-aware bitmap recycling.
- Compress Image
  - Built multi-level image compression (lossy & lossless) with format-specific logic and async execution to maintain a responsive UI.
- Image Background Remover
  - Integrated ML Kit Selfie Segmentation for pixel-level background removal using confidence-based masking, producing clean ARGB images with transparent backgrounds.
- Text-to-Handwriting Generator
  - Developed a handwriting renderer using Canvas, Paint, Typeface, ruled/margined sheet styling, multiple ink colors, and high-resolution A4 (300 DPI) export for aesthetic handwritten outputs.

8 . **POOKIETRAK: A TRACKING & ALERT SYSTEM (6TH SEM)**

3rd July 2025

Technologies: Python, Multi-threading, REST APIs (ISAPI), MySQL, Selenium, AES Encryption.

Developed a comprehensive desktop monitoring solution, "PookieTrak," designed to track software usage patterns and network device availability in real-time. This version established the core architecture for efficient system logging and automated reporting.

- Intelligent Network Scanning: Engineered a high-concurrency scanner (50+ threads) to auto-discover devices and permanently track them using MAC/Serial fingerprinting, ensuring continuity even when IP addresses change.
- Secure Credential Management: Implemented an automated password rotation engine using custom ISAPI integration, ensuring all credentials are AES-512 encrypted and synchronized to a central MariaDB database.
- Enterprise-Grade Alerting System: Developed a robust notification engine using smtplib with auto-recovery, SMTP port heuristics, and detailed HTML reporting to ensure critical infrastructure alerts are delivered reliably (99.9% success rate).
- Hybrid Automation & Reliability: Developed a fault-tolerant architecture that seamlessly falls back to Selenium browser automation for legacy devices, ensuring 100% coverage across different firmware versions.

9 . **BLOCKBEATS: BLOCKCHAIN MUSIC PLAYER (6TH SEM)**

3rd May 2025

Solana-Based Web3 Music Player with Spotify-Inspired UI

- Developed a decentralized music streaming platform to empower emerging artists with low-cost music publishing and distribution
- Engineered smart contracts using Rust with Anchor Framework on Solana blockchain for secure, low-cost music transactions
- Implemented Web3 authentication via Phantom Wallet integration, enabling seamless user sign-in and blockchain interactions
- Created responsive front-end with Next.js/React.js and styled with Tailwind CSS to deliver an intuitive user experience
- Designed dual storage system utilizing Arweave for decentralized permanent storage and Cloudinary for direct URL uploads
- Built robust API infrastructure using Axios to facilitate communication between frontend components and blockchain
- Established RPC connection using Quicknode for reliable blockchain interactions and optimized performance
- Applied modern development practices with VS Code and Postman for efficient testing and debugging

10. BRASS ERP (5TH SEM)

9th January 2025

Developed a modular ERP web application for brass part manufacturing using the MERN stack (MySQL, Express.js, React, Node.js).

- Implemented secure JWT-based authentication and middleware for protected API routes.
- Built RESTful backend APIs for managing clients, quotations, production, sales, and raw materials.
- Designed dynamic, responsive frontend interfaces with React and TailwindCSS.
- Utilized Zustand for efficient global state management across the application.
- Structured backend with MVC architecture, Sequelize ORM for database operations, and clear separation of concerns.
- Integrated robust error handling, input validation, and cookie-based session management.
- Ensured maintainable, scalable code with modular components, reusable utilities, and comprehensive documentation.

11. MOVIE RECOMMENDATION SYSTEM : NETFLIX CLONE (4TH SEM)

20th October 2024

Developed a MERN stack web application designed to replicate core features of Netflix, offering users an immersive experience with a wide variety of movies and genres.

- Engineered a full-stack Netflix-inspired platform using React, Node.js, Express, and MongoDB
- Integrated TMDB API to provide dynamic content including movie details, genres, and recommendations
- Implemented secure user authentication and authorization using JWT with form validation
- Developed personalized features including user profiles, search functionality with history tracking
- Created responsive design ensuring optimal user experience across all devices
- Applied security best practices including encrypted password storage and secure API implementation